

Working with Files in Linux: Sort, Uniq, Diff & WC



sort



What it does

Sorts lines of a file in **ascending order (A–Z)** by default.

```
sort file.txt
```

✓ Important Points

- Sorts alphabetically
- Numbers are sorted lexicographically (unless `-n` is used)
- Does not change original file (just prints output)



sort -r



Reverse sort

Sorts in **descending order (Z–A)**.

```
sort -r file.txt
```



sort -k2



Sort by column (field)

`-k` means **key (column number)**

```
sort -k2 file.txt
```

- Sorts based on **second column**
- Default field separator = space
- Useful for sorting names by last name



Related Important Options

Option	Meaning
<code>-n</code>	Numeric sort
<code>-k</code>	Sort by column
<code>-t</code>	Specify delimiter
<code>-u</code>	Remove duplicates while sorting

Example numeric sort:

```
sort -n marks.txt
```



uniq



What it does

Removes **adjacent duplicate lines**.

```
uniq file.txt
```

⚠ Important:

File must be sorted first for proper duplicate removal.

Correct usage:

```
sort file.txt | uniq
```

uniq -c

◆ **Count duplicate lines**

```
sort file.txt | uniq -c
```

Output example:

```
1 Aarav
```

```
2 Rohan
```

```
1 Priya
```

Shows how many times each line appears.

cmp

◆ **Compare two files byte by byte**

```
cmp file1.txt file2.txt
```

✓ **Output:**

- If same → no output
- If different → shows first difference (line & byte)

Example output:

```
file1.txt file2.txt differ: byte 15, line 3
```

diff

◆ **Compare two files line by line**

```
diff file1.txt file2.txt
```

✓ **Output:**

Shows which lines differ and how.

Example:

```
2c2
```

```
< Hello
```

```
---
```

```
> Hi
```

Means line 2 changed.

◆ **Key Difference**

cmp	diff
Byte comparison	Line comparison
Shows first difference only	Shows all differences
Silent if same	Detailed output

wc (Word Count)

Counts content in a file.

◆ **wc -l**

Counts number of **lines**

```
wc -l file.txt
```

◆ **wc -w**

Counts number of **words**

```
wc -w file.txt
```

◆ **wc -c**

Counts number of **bytes**

```
wc -c file.txt
```

✓ Includes:

- Letters
- Spaces
- Tabs
- Newline characters

📌 **Quick Summary Table**

Command	Purpose
sort	Sort lines A–Z
sort -r	Reverse sort
sort -k2	Sort by 2nd column
uniq	Remove adjacent duplicates
uniq -c	Count duplicates
cmp	Compare files byte-by-byte
diff	Compare files line-by-line
wc -l	Count lines
wc -w	Count words
wc -c	Count bytes